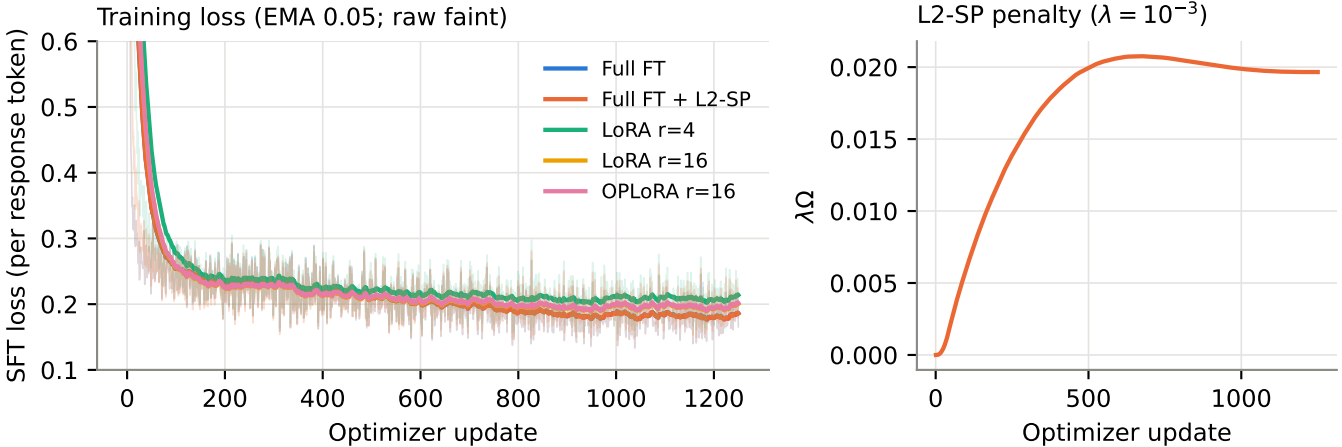

Training loss (EMA 0.05; raw faint)
SFT loss (per response token)
0.6
0.5
0.4
0.3
0.2
0.1
0
200
400
600
800
1000
1200
Optimizer update
Full FT
Full FT + L2-SP
LoRA r=4
LoRA r=16
OPLoRA r=16
L2-SP penalty ($\lambda = 10^{-3}$)
$\lambda\Omega$
0.020
0.015
0.010
0.005
0.000
0
500
1000
Optimizer update